Decimal To Binary Conversion & 1's and 2's Complement:

15

128	64	32	16	8	4	2	1
0	0	0	0	1	1	1	1
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

15-8=7

7-4=3

3-2=1

1-1=0

25

128	64	32	16	8	4	2	1
0	0	0	1	1	0	0	1
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

25-16=9

9-8=1

1-1=0

Decimal to Binary

Binary to Decimal

2

128	64	32	16	8	4	2	1
0	0	0	0	0	0	1	0
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

1+1=2

$$\begin{array}{r} 1 \\ +\ 1 \\ \hline 10 \end{array}$$

3

128	64	32	16	8	4	2	1
0	0	0	0	0	0	1	1
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0

1+1=2

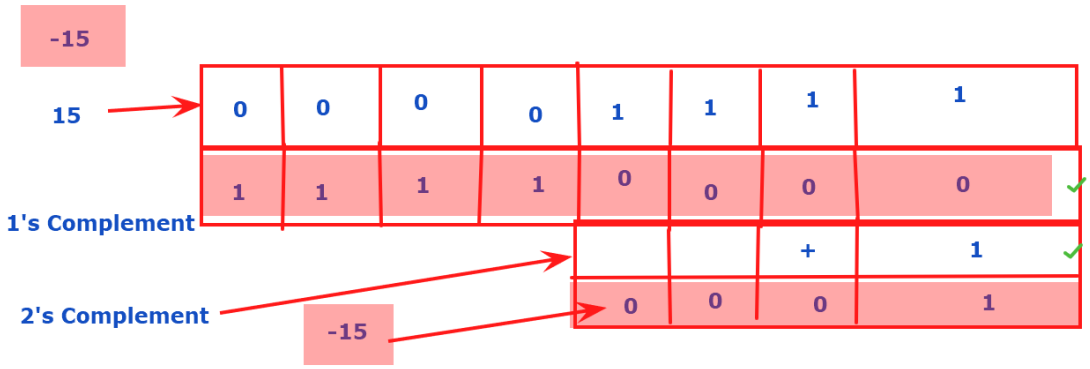
$$\begin{array}{r} 1 \\ +\ 1 \\ \hline 10 \end{array}$$

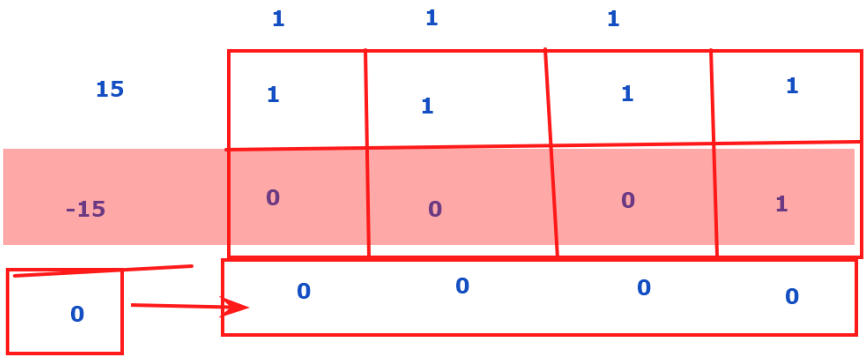
$$\begin{array}{r} 1 \\ 1 \\ +\ 1 \\ \hline 1\ 1 \end{array}$$

1+1+1=3

				1	1	1	1	
5	0	0	0	0	0	1	0	1
15	0	0	0	0	1	1	1	1
20	0	0	0	1	0	1	0	0
		64	32	16	8	4	2	1

-15								
15	0	0	0	0	1	1	1	1
1's Complement	1	1	1	1	0	0	0	0 ✓
						+	1 ✓	
2's Complement					0	0	0	1
-15								


-15
15
0 0 0 0 1 1 1 1
1 1 1 1 0 0 0 0
1's Complement
+ 1
2's Complement
-15
0 0 0 1


1 1 1
15
1 1 1 1
-15
0 0 0 1
0
0 0 0 0